

HAO PAN

Singapore

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Summary

Data Agent / LLM Application Engineer focused on turning real BI workflows into constrained, auditable AI systems that people can use in practice. At Shopee, partnered with product managers and regional stakeholders to deploy production Data Agent Skills for recurring analytics, combining business-logic formalization, deterministic guardrails, reliability evaluation, and human review.

Education

Nanyang Technological University

Nov. 2025 – Jan. 2027

Master of Computing in Applied Artificial Intelligence, College of Computing and Data Science

Singapore

– Coursework: Agentic AI & Applications, Reinforcement Learning, and LLM Applications.

Suzhou University

Sep. 2019 – Jun. 2023

Bachelor of Engineering in Data Science and Big Data Technology

China

– **China National Scholarship** (Top 0.2% nationwide, 2022–2023).

Experience

Shopee Singapore

Mar. 2026 – Sep. 2026

AI Data Analyst Intern, Buyer & Marketing BI Team

Singapore

- Partnered with product managers and regional stakeholders to define business objectives, metric logic, and analytical requirements, then translated recurring A/B-test workflows into **5 production Data Agent Skills** covering PDP, OPC, Review, Cart, and Item.
- Designed a standardized Skill authoring template and codified metric definitions, table relationships, filters, and calculation logic into reusable runbooks; deployed the Skills into actual analytics workflows and reduced turnaround from **30–45 min to 2–3 min**.
- Established deterministic execution and validation controls for reference SQL, parameter handling, shard execution, and result assembly; documented **15 failure modes** and matched manual SQL exactly on **18 rows across all 19 fields** under identical SQL versions and parameters.
- Built validation utilities and an **AI-readable team knowledge base** for historical analyses, business logic, metric definitions, methodologies, and best practices, helping the team build, test, and maintain agent capabilities consistently.

Jinken Vocational and Technical College

Mar. 2023 – Oct. 2025

Instructor & AI Technical Lead

China

- Designed and delivered Python, applied ML, and generative-AI curricula for **300+ students**; created hands-on labs, assessments, and project rubrics.
- Led STEM robotics initiatives and mentored an autonomous indoor-UAV team using ROS, scenario decomposition, and obstacle avoidance to win **Third Prize at the RAICOM 2024 National Finals**.

Selected Projects

Data Agent Sandbox | *TypeScript, React, AlaSQL, Vitest* | [Live Demo](#) | [GitHub](#)

Jul. 2026

- Built a public, browser-only BI Data Agent sandbox that converts business questions into structured contracts, validated SQL, local execution, charts, traces, and downloadable reports using synthetic e-commerce data.
- Implemented deterministic guardrails and a regression-evaluation workspace with testset selection, failure-mode distribution, case-level traces, and a bad-case review queue; runs without login, external services, or an API key.

ContextCue | *TypeScript, Next.js, React* | [Live Demo](#) | [GitHub](#)

Jun. – Jul. 2026

- Built a cross-app context layer for personal AI agents that converts simulated voice instructions and chat/AI-feedback fragments into structured memory and evidence-cited reply context.
- Implemented deterministic MemoryGate guardrails for person/task scoping, human confirmation, TTL-based expiration, and raw chat/audio blocking; authored a **16-case static V0.1 evaluation suite** covering context stitching, wrong-person leakage, privacy, expiry, evidence citation, and tone safety.

Multi-Agent Literature Monitoring System | *Python, Google ADK, ChromaDB, SQLite, Langfuse*

Apr. – May 2026

- Architected the orchestrator for a team-built research-monitoring agent using **SequentialAgent + LoopAgent**, a critic-revision loop, and two human-in-the-loop checkpoints; designed typed schemas, dual storage, tracing, prompts, and guardrails.
- Delivered approximately **3,990 LOC in owned modules** across orchestration, storage, observability, and guardrails; passed **18/18 smoke tests** and the full pytest suite.

Additional build: Clinical Bazi Web | Code-based charting and bounded AI interpretation; private beta, access on request.

Honors

Huawei ICT Competition Global Final | Cloud Track, Global Third Prize (Top 5%)

Jun. 2022

Technical Skills

Languages: Python, SQL (Presto/Trino, Hive, MySQL), C++, Shell

Agent Systems: Google ADK, context engineering, Agent Skills/runbooks, Text-to-SQL, RAG, workflow automation

Reliability & Data: LLM evaluation, failure-mode analysis, guardrails, PyTorch, Transformers, PySpark, SQLite, Langfuse

Engineering: Git, Linux, ROS **Certifications:** Software Designer (PRC), Higher Education Teacher's Certificate in AI (PRC), IELTS 6.5